

BIOLOGY — GRADE 10 | PLANT NUTRITION

Quick Check

Lesson 2: Leaf Structure: The Architecture of a Photosynthetic Machine

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.1: Plant Nutrition | Lesson 2

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In a cross-section of a mango leaf, which layer has tall cells packed closely together just beneath the upper epidermis, each holding many chloroplasts?

A Palisade mesophyll

B Vascular bundle

C Lower epidermis

D Spongy mesophyll

A student examines a mango leaf cross-section under a microscope and looks for stomata with guard cells. Where will she find most of them?

- A** In the lower epidermis
- B** In the upper epidermis, where the sun is strongest
- C** Inside the palisade mesophyll layer
- D** Inside the xylem vessels

The spongy mesophyll of a leaf has many air spaces between its loosely arranged cells. What is the main function of these air spaces?

- A** They absorb sunlight for the palisade cells
- B** They let carbon dioxide diffuse from the stomata to the photosynthesising cells
- C** They store sugars made during photosynthesis
- D** They carry water from the roots to the leaf

A leaf vein (vascular bundle) contains xylem and phloem. Which statement correctly describes their roles?

- A** Both carry only gases in and out of the leaf
- B** Xylem brings water and minerals into the leaf, and phloem carries sugars away from the leaf
- C** Xylem carries sugars away from the leaf, and phloem brings water and minerals in
- D** Xylem absorbs light, and phloem prevents water loss

Kale (sukuma wiki) plants in a Nairobi garden wilt at midday even though the soil is moist, then recover the next morning. Which explanation best fits leaf structure?

- A** In strong sun the stomata open wide, so water vapour escapes faster than the xylem can supply water, and the cells lose turgor
- B** The xylem vessels are permanently damaged by the sun and then repair themselves overnight
- C** The chloroplasts in the palisade cells burst at midday and are replaced at night
- D** The soil loses all its water at midday and gets it back at night

Kale plants moved under the shade of a mango tree stop wilting, but over several weeks the leaves turn pale yellowish-green and the plants grow slowly. Which explanation uses leaf structure correctly?

- A** Shade makes the cuticle thicker, which traps too much water in the leaf
- B** Less light reaches the palisade cells, so less photosynthesis happens, leading to pale leaves and slow growth
- C** Shade blocks all gas exchange because the stomata are sealed shut permanently
- D** Shade makes the xylem carry sugars instead of water

A student revises her model of a sukuma wiki leaf by adding labelled arrows. Which set of arrows is correct?

- A** Light enters through the xylem; water moves up the cuticle; water vapour exits through the palisade cells
- B** Light enters from above through the epidermis; water moves up the xylem; water vapour exits through the stoma
- C** Light enters from below; water moves down the xylem; sugars leave through the stoma
- D** Light enters through the lower epidermis only; water moves up the phloem; oxygen enters the xylem

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).